

Line-Following Robot

PROJECT REPORT · EXAMPLE

Team: [Name Surname], [Name Surname] · Course: [Course] · Date: [Date]

1. Goal

Build a small robot that follows a black line and takes the turns without leaving it.

2. Parts

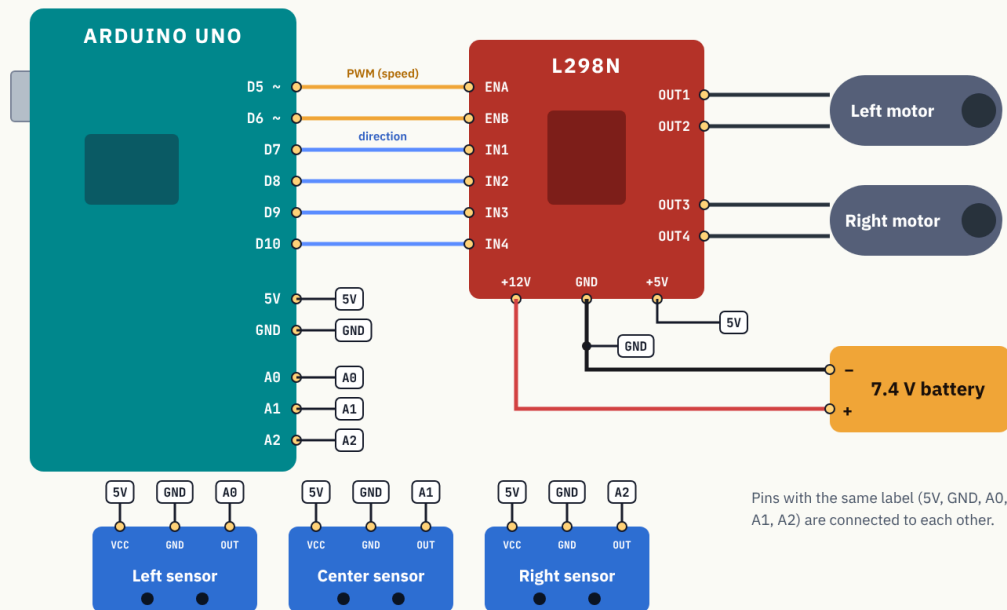
- Arduino Uno
- L298N motor driver
- 2 DC motors with wheels
- 3 infrared line sensors (TCRT5000)
- 7.4 V Li-ion battery

3. How it works

Three sensors read the floor. When the center sensor is on the line, the robot drives straight. When the line drifts left or right, the motor on that side slows down and the robot turns. Speed and the sensor threshold are set at the top of robot.ino.

Circuit diagram: line-following robot

Arduino Uno · L298N motor driver · 3 infrared line sensors



4. Tests

Test	Result	Notes
Straight track	[]	[]
Gentle turn	[]	[]
Sharp turn	[]	[]

5. Files

The code, circuit diagram and photos are in the GitHub repository: [github.com/\[your-username\]/example-project](https://github.com/[your-username]/example-project)

This is an example report made for the class kit; use it as a template for your own project.